MAS508-G Control Theory
Lecture 3: State Space Design

Prof. Jing Zhou

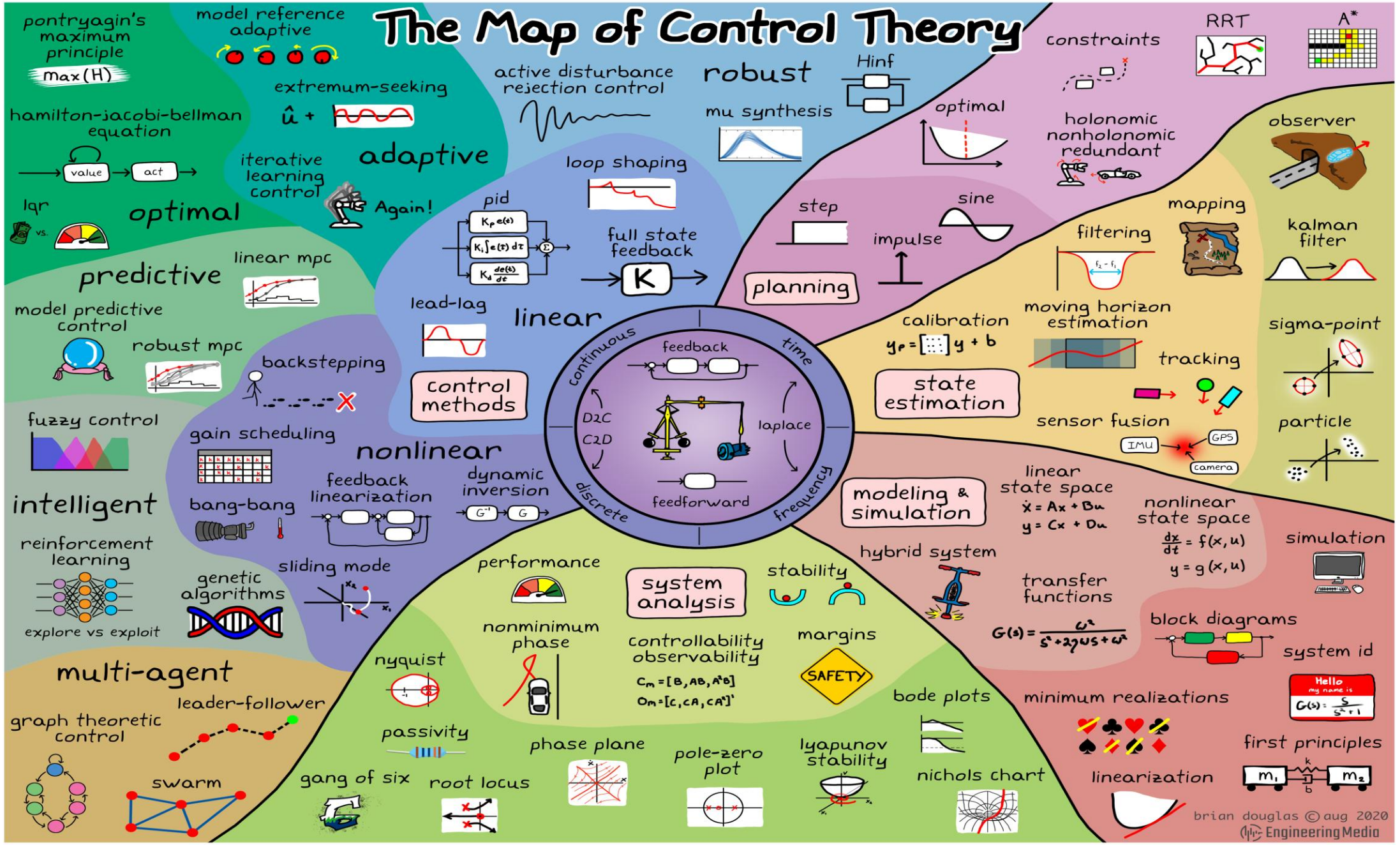
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The Map of Control Theory
pontryagin's maximum principle
Max(H)
model reference adaptive
extremum-seeking
hamilton-jacobi-bellman equation
value
act
iterative learning control
Again!
adaptive
active disturbance rejection control
robust
Hinf
mu synthesis
lqr
vs.
optimal
loop shaping
pid
full state feedback
K
linear mpc
predictive
model predictive control
robust mpc
lead-lag
linear
control methods
backstepping
fuzzy control
gain scheduling
nonlinear
feedback linearization
dynamic inversion
bang-bang
intelligent
reinforcement learning
explore vs exploit
genetic algorithms
sliding mode
multi-agent
graph theoretic control
leader-follower
swarm
continuous
time
discrete
frequency
D2C
C2D
feedback
feedforward
laplace
step
impulse
sine
planning
constraints
RRT
A*
optimal
holonomic
nonholonomic
redundant
observer
filtering
mapping
kalman filter
calibration
$y_p = [\,]\,y + b$
moving horizon estimation
state estimation
tracking
sigma-point
sensor fusion
IMU
GPS
camera
particle
linear state space
$\dot{x} = Ax + Bu$
$y = Cx + Du$
nonlinear state space
$\frac{dx}{dt} = f(x,u)$
$y = g(x,u)$
modeling & simulation
simulation
hybrid system
transfer functions
$G(s) = \frac{\omega^2}{s^2 + 2\zeta\omega s + \omega^2}$
block diagrams
system id
performance
stability
system analysis
nonminimum phase
controllability
observability
$C_m = [B, AB, A^2B]$
$O_m = [C, CA, CA^2]'$
margins
SAFETY
nyquist
passivity
phase plane
pole-zero plot
lyapunov stability
bode plots
nichols chart
gang of six
root locus
minimum realizations
first principles
linearization
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Engineering Media

L3. State Space Design

a) Time Response
b) Controllability
c) Observability
d) Pole Placement Control
e) Observer Design
f) Output Feedback Control
g) Optimal Control - Linear Quadratic Control (LQR)
h) Optimal Control – LQG

Textbook, “Control Systems Engineering”, Chapters 4 & 12

Review of State-Space Model

Book “Control Systems Engineering”, Chapter 3

State-Space Representation

General form:

$$\dot{x} = f(x, u)$$
$$y = h(x, u)$$

Where:

x = vector of state variables $(n \times 1)$

u = the control input vector $(r \times 1)$

y = is the output vector $(m \times 1)$

This is for a **nth order system with *r* inputs and *m* outputs.**

This can describe both *linear* and *nonlinear* systems.

Linear system:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Where:

A = system matrix $(n \times n)$

B = input matrix $(n \times r)$

C = output matrix $(m \times n)$

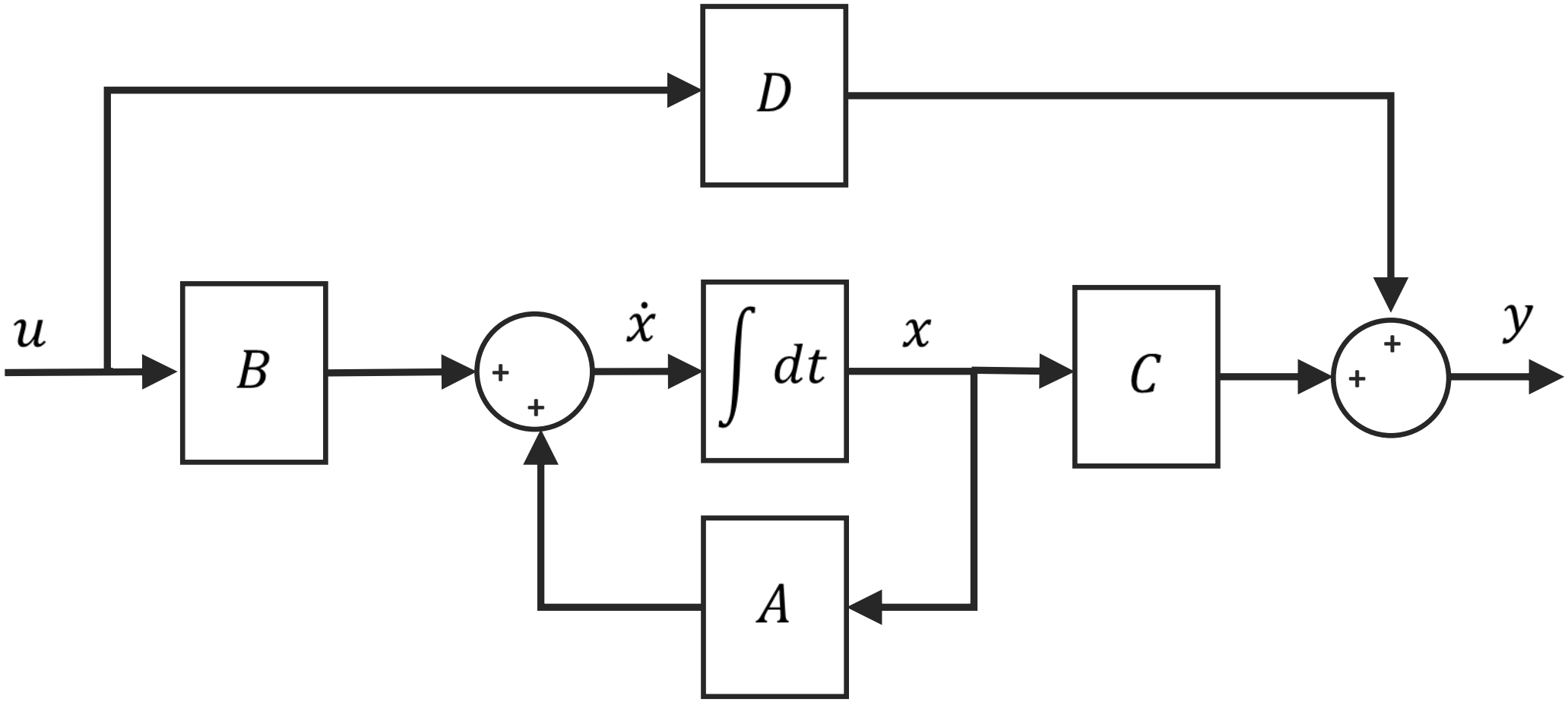
D = feedforward matrix $(m \times r)$

The system state

$$x = [x_1 \quad x_2 \quad x_3 \ldots x_{n-1} \quad x_n]^T$$

State-Space Block Diagram

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$



State-Space Modelling Procedure

1. Find differential equations of system.
2. Define system state vector x $(n \times 1)$, input u $(r \times 1)$, and output y $(m \times 1)$.
3. Write dynamics of system as a set of n first-order equations.
4. Find the matrix A, B, C, D.

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Convert Transfer Function to State-Space

$$y^{(n)} + a_{n-1}\, y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = b_0 u$$

$$\frac{Y(s)}{U(s)} = \frac{b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

Define

$x_1 = y$

$x_2 = \dot{y}$

$x_3 = \ddot{y}$

$\vdots$

$x_{n-1} = y^{(n-2)}$

$x_n = y^{(n-1)}$

$$x = [x_1\ x_2\ \ldots x_n]^T, \quad y = x_1$$

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$$

$$B = [0\ \ 0 \cdots 0\ \ b_0]^T$$

$$C = [1\ 0\ 0 \cdots 0], \qquad D = 0$$

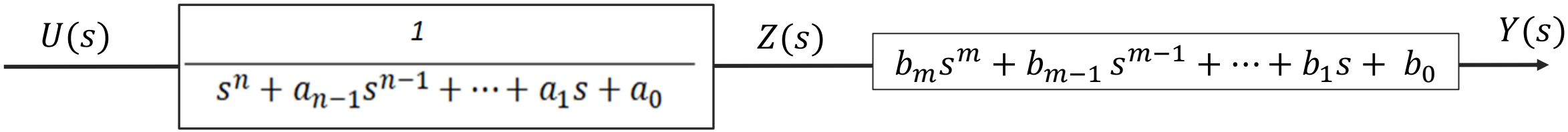
Convert Transfer Function to State-Space

$$y^{(n)} + a_{n-1}\, y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + b_{m-1}\, u^{(m-1)} + \cdots + b_1 \dot{u} + b_0 u, \qquad m < n$$

$$\frac{b_m s^m + b_{m-1}\, s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

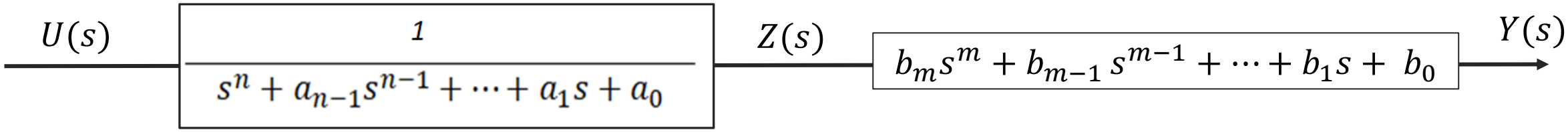
$$\frac{Y(s)}{U(s)} = \frac{Z(s)}{U(s)} \frac{Y(s)}{Z(s)}$$

$$= \left(\frac{1}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0} \right) \cdot (b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0)$$



Convert Transfer Function to State-Space

$$y^{(n)} + a_{n-1}\, y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = b_{n-1} u^{(n-1)} + b_{n-2}\, u^{(n-2)} + \cdots + b_1 \dot{u} + b_0 u$$



$$\frac{Y(s)}{U(s)} = \frac{Z(s)}{U(s)}\frac{Y(s)}{Z(s)} = \left(\frac{1}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}\right)\left(b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0\right)$$

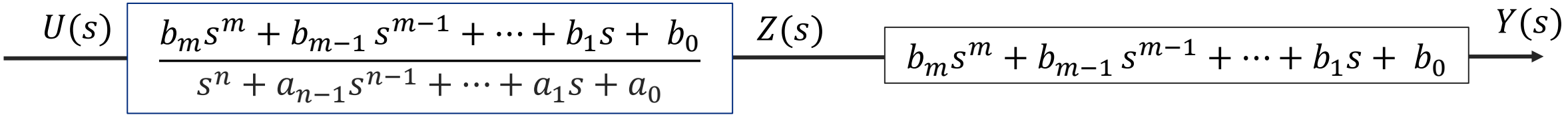
Define $x_1 = z, x_2 = \dot{z}, x_3 = \ddot{z}\,, x_{n-1} = z^{(n-2)}\,, x_n = z^{(n-1)}$

$$x = [x_1\ x_2\ \ldots\ x_n]^T = \left[z\ \ \dot{z}\ \ldots\ z^{(n-2)}\ \ z^{(n-1)}\right]^T$$

$$y = [b_0\ \ b_1\ \ldots.\ b_{m-1}\ \ b_m] \begin{bmatrix} z \\ \dot{z} \\ \vdots \\ z^{(m-1)} \\ z^{(m)} \end{bmatrix}$$

Convert TF to SS: general form

$$y^{(n)} + a_{n-1}\, y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + b_{m-1}\, u^{(m-1)} + \cdots + b_1 \dot{u} + b_0 u$$



$$\frac{Y(s)}{U(s)} = \frac{s^m + b_{m-1}\, s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1}\, s^{n-1} + \cdots + a_1 s + a_0}$$

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [b_0 \;\; b_1 \;\; \ldots \;\; b_{m-1} \;\; b_m \;\; 0_{n-m-1}]\, x$$

Internal state: $x = [x_1 \; x_2 \; \ldots \; x_n]^T = \left[z \;\; \dot{z} \ldots \; z^{(n-2)} \;\; z^{(n-1)}\right]^T$

Convert Transfer function to State-Space

$$y^{(n)} + a_{n-1}\, y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + b_{m-1}\, u^{(m-1)} + \cdots + b_1 \dot{u} + b_0 u$$

Define

$x_1 = z, x_2 = \dot{z}, x_3 = \ddot{z}\,, x_{n-1} = z^{(n-2)}\,, x_n = z^{(n-1)}$

$x = [x_1\ x_2\ \ldots x_n]^T$

$y = [b_0\ \ b_1\ \ldots.\ b_{m-1}\ \ b_m\ \ \ 0_{n-m-1}]\, x$

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$$

$$B = [0\ \ 0 \cdots 0\ \ 1]^T$$
$$C = [b_0\ \ b_1\ \ \ldots.\ b_{m-1}\ b_m\ \ 0_{n-m-1}]$$
$$D = 0$$

Convert State-Space to Transfer Function

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Take Laplace transform for the state space model:

$$SX(s) = AX(s) + BU(s)$$
$$Y(s) = CX(s) + DU(s)$$

Solving:

$$X(s) = (sI - A)^{-1}BU(s)$$

$$Y(s) = C(sI - A)^{-1}BU(s) + DU(s) = [C(sI - A)^{-1}B + D]\ U(s)$$

$$\boxed{T(s) = C(sI - A)^{-1}B + D}$$

Convert State-Space to Transfer Function

State-Space Model:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

$x(n \times 1)$
$y\,(m \times 1)$
$u(r \times 1)$

$A_{n\times n}$
$B_{n\times r}$
$C_{m\times n}$
$D_{m\times r}$

Transfer functions:

$$T(s) = C(sI - A)^{-1}B + D = \begin{bmatrix} \frac{Y_1}{U_1} & \frac{Y_1}{U_2} & \cdots & \frac{Y_1}{U_r} \\ \frac{Y_2}{U_1} & \frac{Y_2}{U_2} & \cdots & \frac{Y_2}{U_r} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{Y_m}{U_1} & \frac{Y_m}{U_2} & \cdots & \frac{Y_m}{U_r} \end{bmatrix}_{m\times r}$$

L3. State Space Design

a) **Time Response**
b) **Controllability**
c) **Pole Placement Control**
d) **Observability**
e) **Observer Design**
f) **Output Feedback Control**
g) **Optimal Control - Linear Quadratic Control (LQR)**
h) **Optimal Control – LQG**

Textbook, “Control Systems Engineering”, Chapters 4 & 12

L3.a Time Response of State-Space

Book “Control Systems Engineering”, Chapter 4.10-4.11

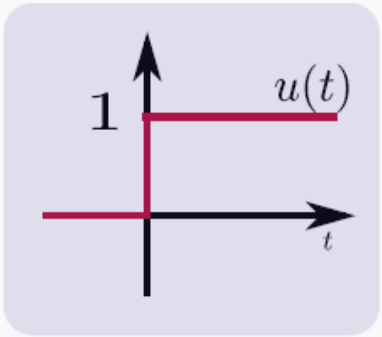
1st-Order Systems

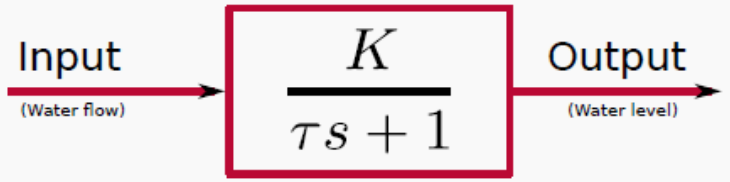
1st-oder Transfer function

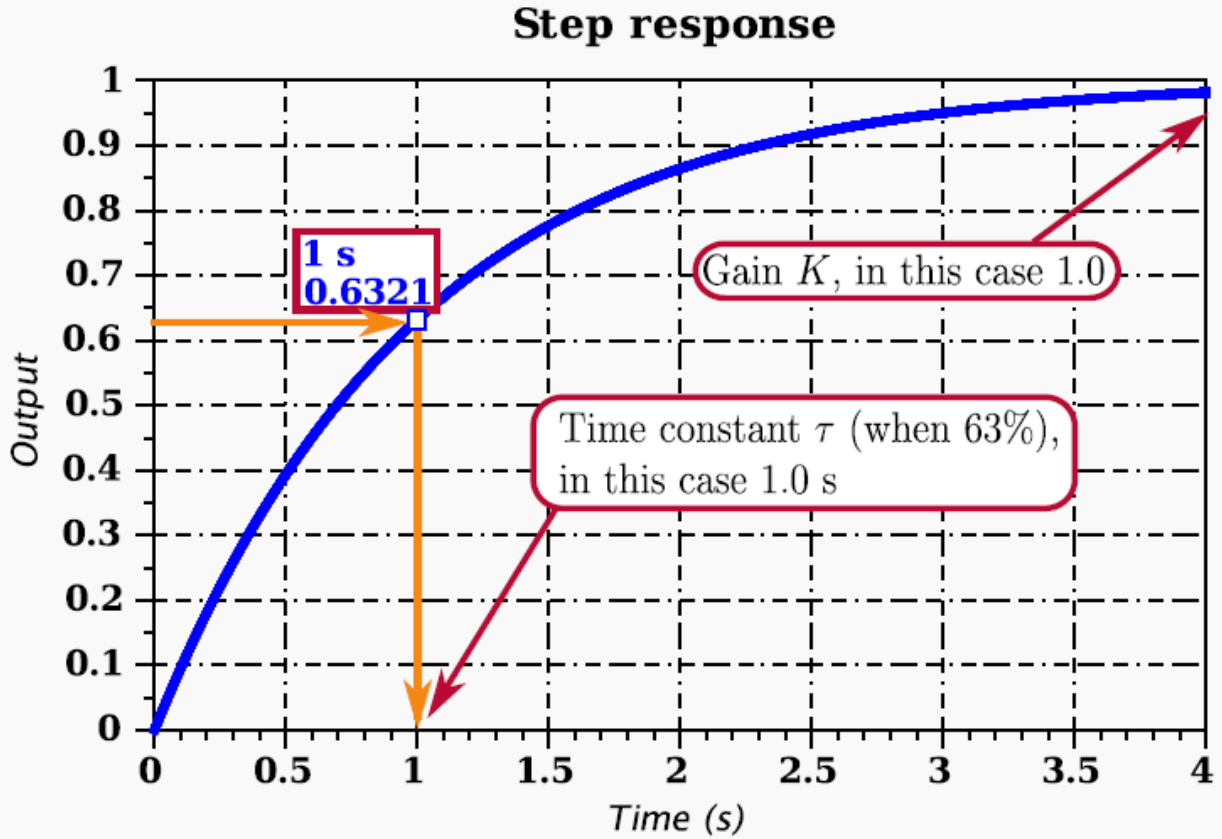
$$G(s) = \frac{K}{\tau s + 1}$$

- K is called the "gain"
- τ is called the "time constant"
- Unit step response: $K\left(1 - e^{-t/\tau}\right)$

- Time Constant τ : Indication of speed of response (convergence) Response is faster as τ becomes smaller.
- Time Constant τ : is the time period in which the step response reaches **63%** of its final value.







Calculate step response

- Transfer function: $G(s) = \frac{C(s)}{U(s)} = \frac{K}{\tau s+1}$
- Unit step response in Laplace domain

$$C(s) = U(s)G(s) = \frac{1}{s}\frac{K}{\tau s+1} = K\left(\frac{1}{s} - \frac{1}{s+1/\tau}\right)$$

- Unit step response:

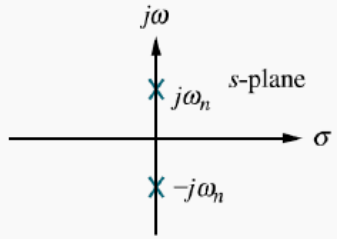
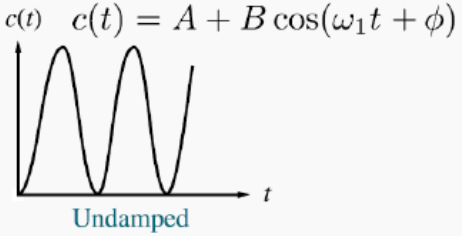
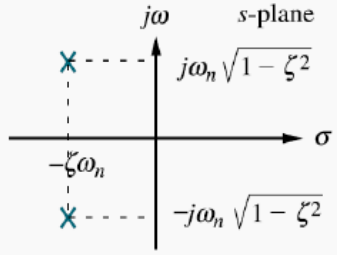
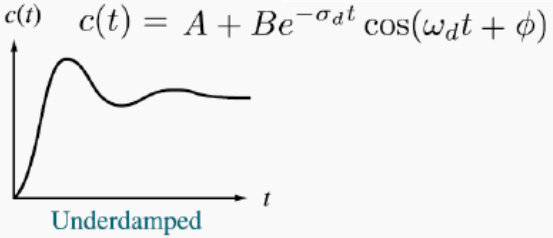
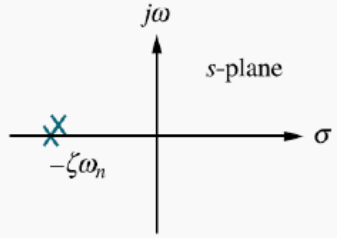
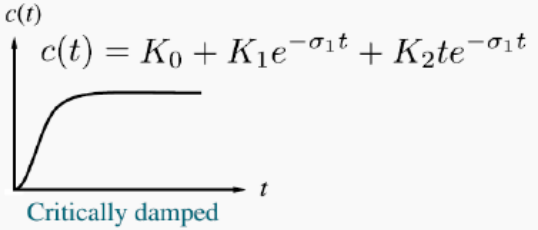
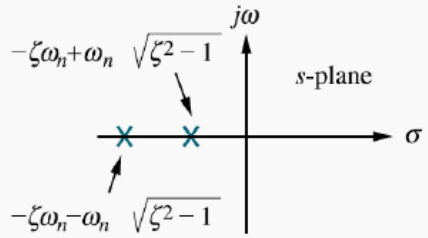
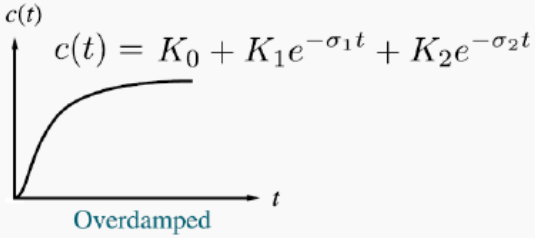
$$c(t) = \mathcal{L}^{-1}[C(s)] = \mathcal{L}^{-1}\left[K\left(\frac{1}{s} - \frac{1}{s+1/\tau}\right)\right] = K(1-e^{-\frac{t}{\tau}})$$

2nd-Order Responses

$$G = \frac{K}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1}$$

Roots:

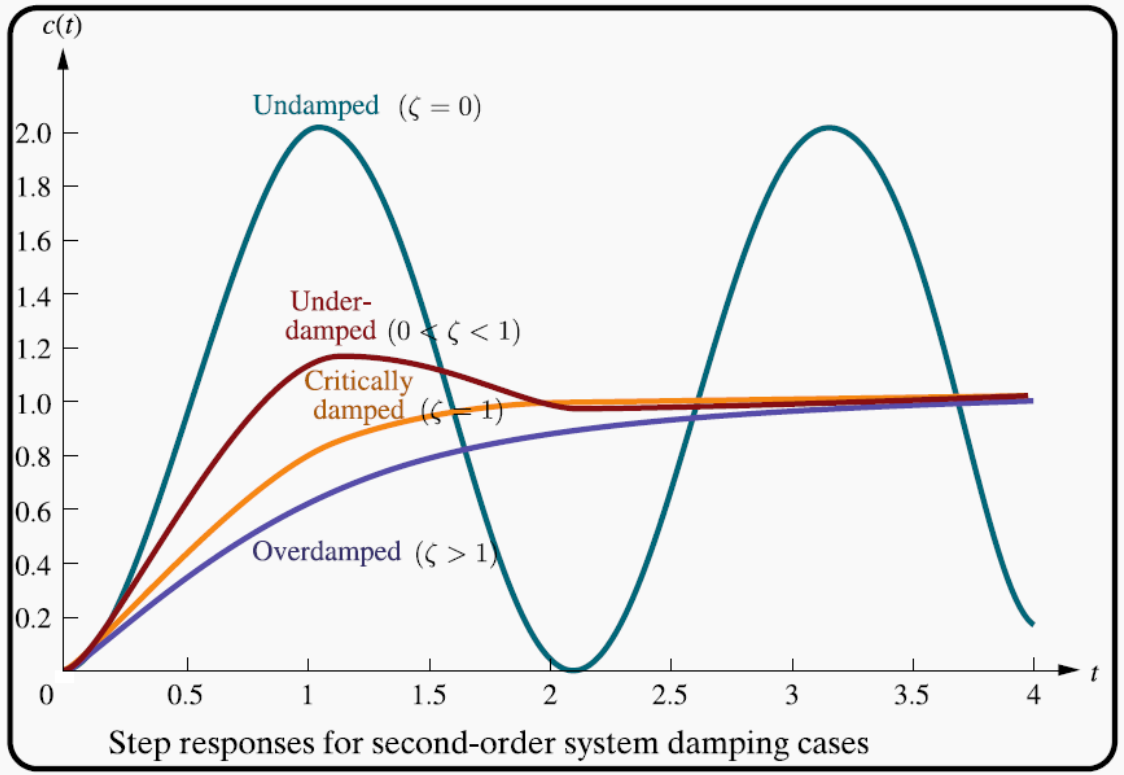
$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

ζ	Poles	Position	Step response
0		Two imaginary at $\pm j\omega_1$	
$0 < \zeta < 1$		Two complex at $-\sigma_d \pm j\omega_d$	
$\zeta = 1$		Two real at $-\sigma_1$	
$\zeta > 1$		Two real at $-\sigma_1, -\sigma_2$	

- Damped natural frequency: frequency at which the system *really* oscillates!

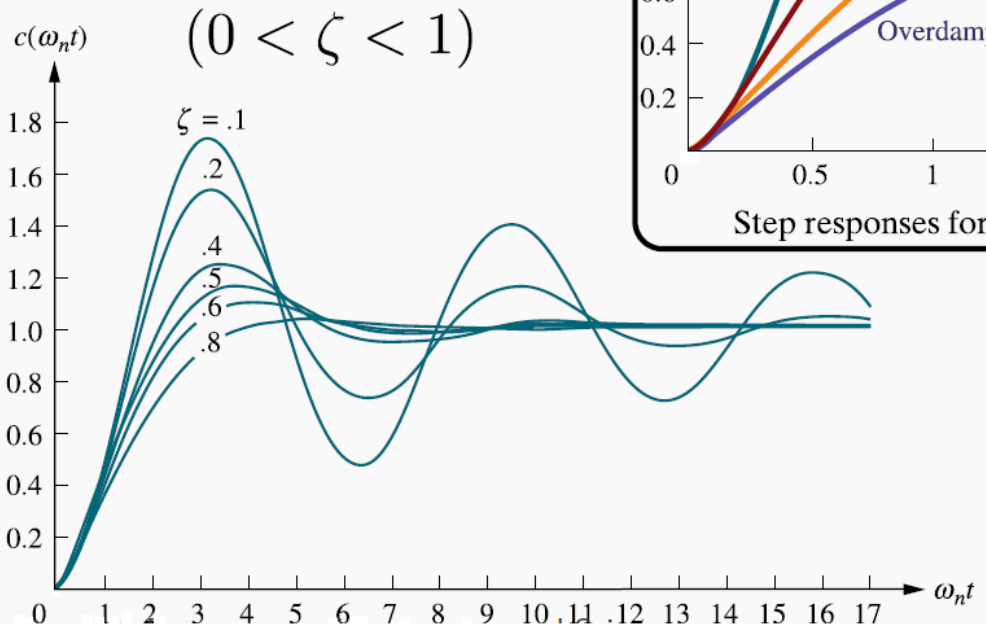
$$\omega_d = \omega_n\sqrt{1 - \zeta^2} \quad (0 < \zeta < 1)$$

2nd-Order Responses

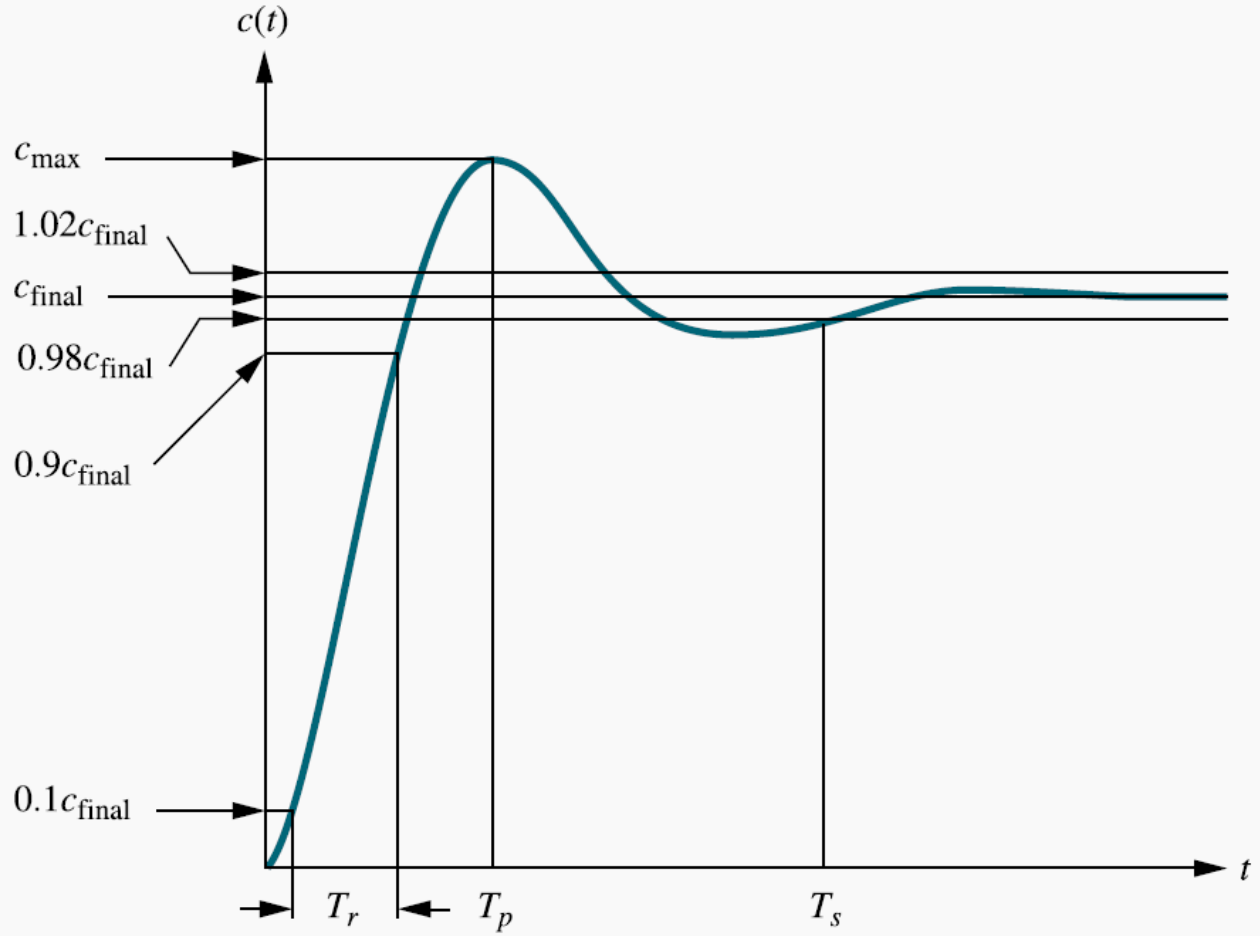


Step responses for second-order system damping cases

2nd order underdamped



2nd-Order Under-Damped Systems



Second-order underdamped response specifications

Rise time: T_r
Time to go from 10% of the final value to 90% of the final value.

Peak time: T_p
Time to reach the first, or maximum peak.

Percent overshoot: $\%OS$
Amount that the waveform overshoots the steady-state value at the peak time. (percentage of the final value: $\frac{c_{max}-c_{final}}{c_{final}}$)

Settling time 2%: T_s
Time to reach and stay within $\pm 2\%$ of the steady-state value.

n^{th}-order system response

- Transfer function: $G(s) = \frac{C(s)}{U(s)} = \frac{K}{(s+p_1)(s+p_2)\ldots(s+p_n)}$
- Unit step response in Laplace domain

$$C(s) = U(s)G(s) = \frac{1}{s}\frac{K}{(s+p_1)(s+p_2)\ldots(s+p_n)} = \frac{K_0}{s} + \frac{K_1}{s+p_1} + \frac{K_2}{s+p_2} + \cdots + \frac{K_n}{s+p_n}$$

- Unit step response:

$$c(t) = \underbrace{K_0}_{\text{Forced response}} + \underbrace{K_1 e^{-p_1 t} + K_2 e^{-p_2 t} \ldots + K_n e^{-p_n t}}_{\text{Natural response}}$$

Laplace transform solution of state equations

- State equation

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

- Take Laplace transform

$$sX(s) - x(0) = AX(s) + BU(s)$$
$$(sI - A)X(s) = x(0) + BU(s)$$

where I is an $n \times n$ matrix, n is the order of the system.

- Solution of state equation is

$$X(s) = (sI - A)^{-1}[x(0) + BU(s)]$$
$$= \frac{\text{adj}(sI - A)}{\det(sI - A)}[x(0) + BU(s)]$$

where adj( ) is the adjoint matrix, det( ) is the determinant of matrix.

Eigenvalues and transfer function poles

- State-space

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

- Transfer function

$$T(s) = C(sI - A)^{-1}B + D$$

$$= C\left[\frac{\text{adj}(sI - A)}{\det(sI - A)}\right]B + D = \frac{C\ \text{adj}(sI - A)B + D\ \det(sI - A)}{\det(sI - A)}$$

- Poles of the system = Roots of denominator = Eigenvalues of system matrix A

$$\det(sI - A) = 0$$

Example (Ex.4.11 in Nise book)

Given the system represented in state-space as following.

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-t} \qquad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$y = [1 \quad 1 \quad 0]\mathbf{x}$$

Do the following:

a) Solve the preceding state equation and obtain the output for the given exponential input.

b) Find the eigenvalues and the system poles.

Solution a)

a) Find the inverse of $(sI - A)$

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 24 & 26 & s+9 \end{bmatrix}$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\begin{bmatrix} (s^2 + 9s + 26) & (s+9) & 1 \\ -24 & s^2 + 9s & s \\ -24s & -(26s + 24) & s^2 \end{bmatrix}}{s^3 + 9s^2 + 26s + 24}$$

Solution a)

$U(s)$ is the Laplace transform for $\mathrm{u}(t) = e^{-t}$, $U(s) = \frac{1}{s+1}$.

$$X(s) = (sI - A)^{-1}[x(0) + BU(s)] = (sI - A)^{-1}\left\{\begin{bmatrix}1\\0\\2\end{bmatrix} + \begin{bmatrix}0\\0\\1\end{bmatrix}\frac{1}{s+1}\right\} = (sI - A)^{-1}\begin{bmatrix}1\\0\\2+\frac{1}{s+1}\end{bmatrix}$$

$$X_1(s) = \frac{(s^3 + 10s^2 + 37s + 29)}{(s+1)(s+2)(s+3)(s+4)}$$

$$X_2(s) = \frac{(2s^2 - 21s - 24)}{(s+1)(s+2)(s+3)(s+4)}$$

$$X_3(s) = \frac{s(2s^2 - 21s - 24)}{(s+1)(s+2)(s+3)(s+4)}$$

Solution a)

The output is

$$Y(s) = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_1(s) \\ X_2(s) \\ X_3(s) \end{bmatrix} = X_1(s) + X_2(s) = \frac{(s^3 + 12s^2 + 16s + 5)}{(s+1)(s+2)(s+3)(s+4)}$$

$$= \frac{-6.5}{s+2} + \frac{19}{s+3} - \frac{11.5}{s+4}$$

Taking the inverse Laplace transform of $Y(s)$, the output is

$$y(t) = -6.5e^{-2t} + 19e^{-3t} - 11.5e^{-4t}$$

Solution b)

b) The roots of denominator of $(sI - A)^{-1}$ is $\det(sI - A)$.

$$\det(sI - A) = s^3 + 9s^2 + 26s + 24 = (s + 2)(s + 3)(s + 4) = 0$$

Thus, the poles of system and the eigenvalues are -2, -3, and -4.

Time domain solution of State equations

- State equation

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

- Laplace domain solution

$$X(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s)$$

- Time domain solution

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}B\,u(\tau)d\tau$$
$$= \Phi(t)x(0) + \int_0^t \Phi(t-\tau)B\,u(\tau)d\tau$$

- $\Phi(t) = e^{At}$ is called the state-transition matrix. $(sI - A)^{-1}$ is the Laplace transform of $\Phi(t)$, or $\mathcal{L}^{-1}[(sI - A)^{-1}] = \Phi(t)$

Example

Find a) the state-transition matrix and b) unit step response $x(t)$ of the following state equation.

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Solution a)

- Using the inverse Laplace transform method $\Phi(t) = \mathcal{L}^{-1}[(sI - A)^{-1}]$.
- Find $(sI - A)$ as

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & -1 \\ 8 & (s+6) \end{bmatrix}$$

- Find its inverse

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\begin{bmatrix} s+6 & 1 \\ -8 & s \end{bmatrix}}{s^2+6s+8} = \begin{bmatrix} \frac{s+6}{s^2+6s+8} & \frac{1}{s^2+6s+8} \\ \frac{-8}{s^2+6s+8} & \frac{s}{s^2+6s+8} \end{bmatrix} = \begin{bmatrix} \left(\frac{2}{s+2} - \frac{1}{s+4}\right) & \left(\frac{1/2}{s+2} - \frac{1/2}{s+4}\right) \\ \left(\frac{-4}{s+2} + \frac{4}{s+4}\right) & \left(\frac{-1}{s+2} + \frac{2}{s+4}\right) \end{bmatrix}$$

- Finally, taking the inverse Laplace transform of each term, we obtain

$$\Phi(t) = \mathcal{L}^{-1}[(sI - A)^{-1}] = \begin{bmatrix} (2e^{-2t} - e^{-4t}) & \left(\frac{1}{2}e^{-2t} - \frac{1}{2}e^{-4t}\right) \\ (-4e^{-2t} + 4e^{-4t}) & (-e^{-2t} + 2e^{-4t}) \end{bmatrix}$$

Solution b) Unit step response with initial values

- Using the inverse Laplace transform method for unit step input $U(s) = \frac{1}{s}$

$$X(s) = (sI - A)^{-1}[x(0) + BU(s)] = \begin{bmatrix} \frac{s+6}{s^2+6s+8} & \frac{1}{s^2+6s+8} \\ \frac{-8}{s^2+6s+8} & \frac{s}{s^2+6s+8} \end{bmatrix} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{s} \right)$$

$$= \begin{bmatrix} \frac{1}{8s} + \frac{7}{4(s+2)} - \frac{7}{8(s+4)} \\ -\frac{7}{2(s+2)} + \frac{7}{2(s+4)} \end{bmatrix}$$

- Taking the inverse Laplace transform, we obtain

$$x(t) = \begin{bmatrix} \frac{1}{8} + \frac{7}{4}e^{-2t} - \frac{7}{8}e^{-4t} \\ -\frac{7}{2}e^{-2t} + \frac{7}{2}e^{-4t} \end{bmatrix}$$

```
% Step response with state-space model
A = [0 1; -8 -6];
B=[0; 1;];
C=[1 0; 0 1];
D=0;
x0=[1; 0];
sys = ss(A,B,C,D);
t=0:0.1:10;
[x,t]=step(sys,t);
[xi,t]=initial(sys,x0,t);
figure(1);plot(t,x+xi);title('Step response with initials')
```

Takeaways

Benefits

- Easier to evaluate Multiple-Input-Multiple-Output time response using state-space approach.

Drawbacks

- Not necessary for SISO and other simple system, transfer function method is easier to use.